Hieu Tran

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Research Interests

Mechanistic Interpretability & Steering: representation engineering, activation steering, causal interventions
Multilingual LLMs & Reasoning: cross-lingual transfer, multilingual reasoning, synthetic data augmentation
LLM Agents & Multi-Agent Systems: autonomous agents, deep research, coding agents, tool use optimization

Education

University of Maryland, College Park — Maryland, USA
PhD in Computer Science, supervised by Prof. Marine Carpuat — *Aug. 2024 – Expected May 2029*
- Achievements: Dean's Scholarship

VNUHCM - University of Science — Ho Chi Minh, Vietnam
Bachelor of Computer Science — *Aug. 2017 – Jul. 2021*
- Achievements: Outstanding Undergraduate Thesis Award

Research Experience

Graduate Research Assistant — Aug. 2024 – Present
University of Maryland, College Park — *Maryland, USA*
- Investigate mechanistic interpretability and controllable language generation through activation steering, representation analysis, and causal interventions.
- Explore multilingual LLM reasoning through internal-state probing to reduce redundant thinking tokens and improve test-time compute efficiency.
- Develop multilingual NLP methods for cross-lingual transfer and synthetic data augmentation.

Selected Publications

1. **Hieu Tran**, Sy-Tuyen Ho, Marine Carpuat. 2026. What Does Steering Change? Decoding with Base-Relative Steering Evidence. *Submitted to ICLR 2027* — Paper

2. Tai Dang, **Hieu Tran**, Sang T. Truong, Long-Hung Pham, Edward A Pham, Jeffrey S. Glenn, Thang Luong. 2026. Plausible Biomolecular Structure Prediction via Physics-informed Reinforcement Learning. *NeurIPS 2026*

3. **Hieu Tran**, Calvin Bao, Marine Carpuat. 2026. Can Activation Steering Capture Multidimensional Authorship Style?. In *EMNLP 2026* — Paper

4. **Hieu Tran***, Phuong-Anh Nguyen-Le*, Huy Nghiem, Quang-Nhan Nguyen, Wei Ai, Marine Carpuat. 2026. VietMix: A Naturally Occurring Vietnamese-English Code-Mixed Corpus with Iterative Augmentation for Machine Translation. In *EACL 2026* (Oral) — Paper

5. Long Phan*, Tai Dang*, **Hieu Tran***, Trieu H. Trinh*, Vy Phan, Lam Chau, Thang Luong. 2023. Enriching Biomedical Knowledge for Low-resource Language Through Large-Scale Translation. In *EACL 2023* — Paper

6. Chinh Ngo*, Trieu H. Trinh*, Long Phan*, **Hieu Tran***, Hieu Nguyen, Tai Dang, Minh Nguyen, Minh-Thang Luong. 2022. MTet: Multi-domain Translation for English and Vietnamese. In *ArXiv Preprint* — Paper

7. Long Phan, **Hieu Tran**, Hieu Nguyen and Trieu H. Trinh. 2022. ViT5: Pretrained Text-to-Text Transformer for Vietnamese Language Generation. In *NAACL SRW 2022* — Paper

8. **Hieu Tran**, Long Phan, James T. Anibal, Binh T. Nguyen and Truong-Son Nguyen. 2021. SPBERT: An Efficient Pre-training BERT on SPARQL Queries for Question Answering over Knowledge Graphs. In *ICONIP 2021* — Paper

9. Long Phan, **Hieu Tran**, Daniel Le, Hieu Nguyen, James T. Anibal, Alec Peltekian and Yanfang Ye. 2021. CoTexT: Multi-task Learning with Code-Text Transformer. In *NLP4Prog @ ACL-IJCNLP 2021* — Paper

Industry Experience

Senior Machine Learning Engineer Jun. 2023 – Jul. 2024
BLACKBOX AI *Remote (Canada)*

- Pioneered core features as an early engineer for flagship products, including the VSCode Coding Agent, Web Agent, Auto Completion, scaling the user base from their early stages to over **30M users** and **4M** VSCode downloads
- Owned the end-to-end development of the code search/web search system, from researching ranking algorithms to implementing the final retrieval pipeline used in production
- Trained CodeLlama-34B on synthetic instruction-following datasets, achieving **86.6%** on HumanEval, surpassing GPT-4 (Nov 2023) which scored 85.4%

Lecturer & Research Collaborator May. 2021 – Aug. 2024
VietAI *Ho Chi Minh, Vietnam*

- Taught Foundations of Deep Learning, Advanced NLP, and LLMs courses to over **200 students**
- Worked directly with Dr. Minh-Thang Luong and Dr. Trieu H. Trinh (Google DeepMind) to build English-Vietnamese machine translation systems that outperform previous SOTA by 4.5% (BLEU).
- Released the first Vietnamese generative models, achieving SOTA in Summarization and Biomedical benchmarks.

AI Research Engineer Aug. 2021 – Aug. 2023
M_Service (MoMo) (3rd unicorn in Vietnam) *Ho Chi Minh, Vietnam*

- Built Intent-based/ChatGPT-based AI chatbots for Customer Support on MoMo #1 e-wallet platform in Vietnam
- Managed the training pipeline and chatbots on Kubernetes deployed on Google Cloud Platform
- Trained base language models that outperformed open-source Vietnamese models for our internal needs

AI Software Engineering Intern Apr. 2019 – Aug. 2021
Zalo Group - VNG Corporation (1st unicorn in Vietnam) *Ho Chi Minh, Vietnam*

- Constructed RDF-triplestore Knowledge Bases for Kiki - the first Vietnamese AI virtual assistant
- Proposed a light and state-of-the-art Vietnamese Spelling Corrector.

Projects

Coding Agent on Browser | *TypeScript, APIs, CodeSandbox*

- Built a browser UI for user-submitted coding tasks; an LLM agent follows the *ReAct* framework to plan steps and decide which tools to call
- Executes a chain of tools to operate on the project file system, edit/read/search files and run a sandboxed terminal, to iteratively complete tasks end-to-end

Multi-Agent Machine Translation System | *Python, Transformers, LiteLLM*

- Designed a multi-agent framework that orchestrates specialized LLM agents (*ResearchAgent, DraftAgent, RefineAgent, EvaluationAgent*) to collaboratively perform translation for low-resource languages

Deep Research Agent System | *Python, LLM Agents, RAG*

- Developed a deep research agent that autonomously plans, retrieves, and synthesizes information from the web and internal knowledge sources to produce comprehensive, citation-grounded research reports
- Implemented multi-step reasoning, tool-use orchestration, and iterative refinement to improve factuality, coverage, and reliability of long-horizon research tasks

Technical Skills

Programming: Python, Kotlin, TypeScript, SQL
ML Frameworks: PyTorch, Transformers, DeepSpeed, PEFT, scikit-learn
Infrastructure: Kubernetes, Docker, Google Cloud Platform, Elasticsearch, PostgreSQL, MongoDB
LLM Training & Inference: Distributed Training, LLM Fine-tuning, vLLM, Model Quantization, KV-Cache Optimization, Distributed Inference